UM-2 provenance

Where every ingredient, equation and number of the universe model UM-2 comes from, how it was computed, and how it was verified

Underlying theory: Patrick L. Nash Derived and verified by Claude (Anthropic)

Provenance, version 1, 2026-10-17

Repository once-ere/Dirac_claude, folder universe_model_2

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1 How to read this document

This document records the provenance of UM-2, a model of the Universe built on the author's 4+4-dimensional Einstein-Lovelock theory and his dirac16complex field. For every ingredient it states where it comes from, writes the ingredient out as an equation, names the computation that produced or used it, and states how it was verified. The document is self-contained: every equation it mentions is written here.

Labels used throughout:

- **AUTHOR**: taken from the author's Mathematica notebooks or the author's written orders;
- **RECORD**: taken from the Revision record of this repository (earlier work carried out under the author's orders, with its own verification);
- **STANDARD**: textbook physics or a measured value; measured values are marked *verified* (read from the source on 2026-10-17) or *from memory* (quoted from memory of the standard tables, to be checked against the current editions);
- **ASSUMED**: a modelling choice made for UM-2;
- **INVENTED**: a new phenomenological interaction of UM-2;
- **DERIVED**: computed and verified in UM-2;
- **ESTIMATE**: an order-of-magnitude evaluation, not a precision computation;
- **OPEN**: not resolved in this version.

Units: $\hbar = c = 1$; H is the author's constant (an inverse length); in the numerical work $H = 1$ unless a physical unit is stated.

2 The order

The model answers the author's order of 2026-10-17. Its key parts, in the author's words: "Create a new model of the Universe that includes [a] gravity (Einstein-Lovelock gravity); [b] the dirac16complex quantized spinor field (which is hypothesied to represent dark matter); the dirac16complex is coupled to gravity using a non-trivial canonical spin connection; ... interactions of gravity and dirac16complex with SOME of the following Standard Model particles: [c] photons; [d] neutrinos; [e] electrons; [f] quarks (only up and down, for the protons and neutrons); [g] the 3 weak interaction gauge bosons, the W and Z 'particles'. However do NOT include the Higgs particle. You must invent/create new phenomenological interactions ... consistent with qcd and electro-weak models. You want to be able to model: CMB, CNB, BAO, so your model for the metric tensor ... must be very general, but perhaps expressible as a power series of terms of decreasing magnitude. Put each and every interaction in equation form, and explain and analyze it in documentation and provenance files; these should be both markdown and latex file types; compile all latex to pdf. NEVER refer the reader to some other file! PUT EVERY EQUATION you create and/or use IN THESE FILES. CRITICALLY IMPORTANT: Mass will be given to all non-massless particles, (except the massless particles: graviton, photon and perhaps the quanta of the dirac16complex field), by appropriate wave function dependence on the space coordinate x0 ... periodic in x0 with period $\pi/(6H)$. Localization with respect to x0 implies, using Heisenberg ... $\Delta x_0\, \Delta p_{x_0} \geq \hbar/2$ ... [2] push your new, improved, model ... along with provenance (both .md and .pdf files). [3] Compute the coupled field equations and push to repo; create concomitant documention (both .md and .pdf files) and push. [4] Compute an accurate approximate solution to the coupled field equations (use Kohn-Sham theory for the dirac16complex field, which couples to all other fields) and push ... Check and verify this repo."

Further orders of the same day: "analyze the x0 mass mechanism, and ... analyze the validity of associating dark matter (and dark energy) with dirac16complex"; "the W^+ and W^- bosons carry positive and negative electric charges. Do not forget this"; the coordinates are always the author's $\{x_0, x_1, \ldots, x_7\}$ of the notebook `Dirac_Einstein-Lovelock_4+4_Nash.nb`, never relabelled; every verified result is pushed to the repository as soon as it is obtained; no single calculation runs longer than 7 minutes and 3 seconds (423 s).

3 Who did what

- The 4+4 framework, the coordinates, the primordial metric of the notebooks, the Einstein-Lovelock action with its weights and Λ, the dirac16complex field and its $Cl(4,4)$ gamma matrices, the mass parameter M, the idea that masses come from the x_0 dependence of wave functions periodic with period $\pi/(6H)$, and the dark-matter hypothesis are the AUTHOR's (Patrick L. Nash).
- The corrected primordial metric (version 2), the regular tip cutoff, the Dirac-type Lagrangian with the canonical spin connection, the 2×2 block reduction of the Kohn-Sham problem and its conventions are RECORD items: produced earlier in this repository under the author's orders and verified there.
- Everything labelled ASSUMED, INVENTED or DERIVED below, all the code of UM-2 and all its documents are by Claude (Anthropic), working in this repository on 2026-10-17.

4 Ingredients from the author (AUTHOR)

4.1 Coordinates and signature

The author's coordinates (notebook `Dirac_Einstein-Lovelock_4+4_Nash.nb`, `X = {x0[], x1[], ..., x7[]}`) are

$$X = \{x_0, x_1, x_2, x_3, x_4, x_5, x_6, x_7\} :$$

x_0 a hidden space-like direction, x_1, x_2, x_3 ordinary space, x_4 the time, x_5, x_6, x_7 three extra time-like directions. The orthonormal frame metric is

$$\eta_{ab} = \mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1), \qquad a,b = 0,\dots,7 \text{ aligned with } x_0,\dots,x_7.$$

The hidden coordinate enters through $z = 6Hx_0$; one period of x_0 is $\ell_0 = \pi/(6H)$.

4.2 The primordial metric of the notebooks

The notebook `WaveFunctionUniverse-4+4-Einstein-Lovelock.nb` (the author's local notebook) gives the primordial metric

$$g^{\mathrm{nb}} = \mathrm{diag}\big(\cot^2 z,\ e^{2a_4}(\sin^2 z)^{1/6},\ e^{2a_4}(\sin^2 z)^{1/6},\ e^{2a_4}(\sin^2 z)^{1/6},\\ -1,\ -e^{-2a_4}(\sin^2 z)^{1/6},\ -e^{-2a_4}(\sin^2 z)^{1/6},\ -e^{-2a_4}(\sin^2 z)^{1/6}\big),$$

with vielbein $\mathrm{diag}(\cot z, e^{a_4}(\sin^2 z)^{1/12} \times 3, 1, e^{-a_4}(\sin^2 z)^{1/12} \times 3)$ and $a_4 = a_4(x_4)$. By the author's orders of 2026-10-14 a second function $b_4(x_4) > 0$ multiplies the extra-time components ("b4 is a function of x4; b4 is NEVER 1"), and $(\sin z)^{1/3}$ means $(\sin^2 z)^{1/6}$, the positive branch, for every real z.

4.3 The Einstein-Lovelock action, weights and cosmological term

The notebook writes the Einstein-Lovelock vacuum equations (after Lovelock and Rund) as

$$0 = -\Lambda^{\mathrm{nb}} + H^{-2} w_1\, \mathrm{Lovelock1} + H^{-4} w_2\, \mathrm{Lovelock2} + H^{-6} w_3\, \mathrm{Lovelock3},$$

with $(w_1, w_2, w_3, \Lambda^{\mathrm{nb}}) = (1, 1/32, 1/1152, 10)$: in the notebook `Dirac_Einstein-Lovelock_4+4_Nash.nb` (In[133] to In[136]) the author sets $w_1 = 1$, $w_2 = 1/2^5$ and solves the vacuum equations on the history $a_4 = Hx_4$, $b_4 = 1$ of the notebook metric for w_3 and Λ^{nb} (In[134]); In[136] factors $1152 = 2^7\, 3^2$; the Revision specification records the solution $w_3 = 1/1152$, $\Lambda^{\mathrm{nb}} = 10$. Multiplying by H^2 and writing the equations as $\sum_k \alpha_k {E_{(k)}}^M{}_N + \Lambda\, \delta^M_N = \kappa T^M{}_N$ gives the UM-2 form

$$\alpha_k = w_k H^{2-2k}, \qquad w = \Big(1, \frac{1}{32}, \frac{1}{1152}\Big), \qquad \Lambda = -\Lambda^{\mathrm{nb}} H^2 = -10H^2,$$

where

$${E_{(k)}}^M{}_N = -\frac{1}{2^{k+1}}\, \delta^{M A_1 B_1 \cdots A_k B_k}_{N C_1 D_1 \cdots C_k D_k}\, R^{C_1 D_1}{}_{A_1 B_1} \cdots R^{C_k D_k}{}_{A_k B_k},$$

$$\mathcal{L}_{(k)} = \frac{1}{2^k}\, \delta^{A_1 B_1 \cdots A_k B_k}_{C_1 D_1 \cdots C_k D_k}\, R^{C_1 D_1}{}_{A_1 B_1} \cdots R^{C_k D_k}{}_{A_k B_k},$$

$\delta^{\cdots}_{\cdots}$ being the generalised Kronecker delta (the determinant of ordinary deltas); $E_{(1)}$ is the Einstein tensor, $\mathcal{L}_{(1)} = R$, $\mathcal{L}_{(2)} = R^2 - 4R_{AB}R^{AB} + R_{ABCD}R^{ABCD}$. That the UM-2 normalisation reproduces the record's equations (which were checked against the notebook) is verified exactly (section 8, script `um2_geometry.py`, checks `record_equation`). UM-2 keeps the author's weights and Λ unchanged; an earlier model of this repository (UM-1) had changed w_3 and Λ.

4.4 The gamma matrices of $Cl(4,4)$

The author's notebooks (*Dirac_Einstein-Lovelock_4+4_Nash.nb* and *WaveFunctionUniverse-4+4-Einstein-Lovelock.nb*) build real 16×16 matrices from ($h = 1,2,3$; $p,q = 1,\ldots,4$; ϵ the Levi-Civita symbol)

$$Q_a(h,p,q) = \epsilon_{hpq4}, \qquad Q_b(h,p,q) = \delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4}, \qquad s_h = Q_a - Q_b, \quad t_h = Q_a + Q_b,$$

$$\zeta_0 = I_8, \quad \zeta_{7-h} = \begin{pmatrix} 0 & t_h \\ -t_h & 0 \end{pmatrix}, \quad \zeta_h = \begin{pmatrix} 0 & s_h \\ s_h & 0 \end{pmatrix}, \quad \zeta_7 = \zeta_1\zeta_2\zeta_3\zeta_4\zeta_5\zeta_6, \quad \sigma = \begin{pmatrix} 0 & I_4 \\ I_4 & 0 \end{pmatrix},$$

$$\tau_0 = I_8, \quad \tau_A = \zeta_A, \quad \bar\tau_A = \sigma\,\tau_A^T\,\sigma, \qquad \gamma^{x_A} \equiv T16[A] = \begin{pmatrix} 0 & \bar\tau_A \\ \tau_A & 0 \end{pmatrix} \quad (A = 0,\ldots,7).$$

They satisfy $\{\gamma^a,\gamma^b\} = 2\eta^{ab}I_{16}$ and the notebook identities $\frac{1}{2}(\tau_A\bar\tau_B + \tau_B\bar\tau_A) = \eta_{AB}I_8$, $\sigma = \tau_1\tau_2\tau_3 = \tau_4\tau_5\tau_6\tau_7$ (notebook Out[322] to Out[324]). Derived objects used by UM-2:

$$C = \gamma^{x_0}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3} = \begin{pmatrix} -\sigma & 0 \\ 0 & \sigma \end{pmatrix}, \qquad \Gamma = \gamma^{x_0}\cdots\gamma^{x_7} = \mathrm{diag}(-I_8, I_8), \qquad B = -iC\gamma^{x_4}, \qquad S^{ab} = \tfrac{1}{4}[\gamma^a,\gamma^b].$$

4.5 The mass parameter, the x_0 mass idea and the dark-matter hypothesis

The dirac16complex mass is written $m = 3HM$ in the author's notebooks. The order requires every non-massless particle to get its mass from the dependence of its wave function on x_0, periodic with period $\pi/(6H)$, with the uncertainty relation $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$ as the physical reason, and states the hypothesis that the quanta of dirac16complex represent dark matter (and asks for the validity of a dark-energy association to be analysed).

5 Ingredients from the Revision record (RECORD)

5.1 The corrected primordial metric (version 2)

The author's order of 2026-10-16: "a diagonal form for the metric of the primordial gravitational field is only self-consistent if all off-diagonal terms of all source terms in the Einstein field equations ARE ZERO. This is not the case here. I task you to: [1] find the simplest modification for the current metric of the primordial gravitational field that is probably self-consistent with the Einstein field equations". The metric of 2026-10-14 (the notebook metric with b_4) has the off-diagonal component $E_{(k)}{}^{x_4}{}_{x_0} \propto \cot z\,\dot b_4/b_4$ at every Lovelock order. In the family (with $g_{44} = -1$)

$$g_{00} = b_4^{2p}e^{2qa_4}\cot^2 z, \qquad g_{ii} = e^{2a_4}(\sin^2 z)^{3\alpha}, \qquad g_{tt} = -e^{-2a_4}b_4(\sin^2 z)^{3\beta}$$

the mixed curvature is proportional to $\alpha\big((q-1)\dot a_4 + p\,\dot b_4/b_4\big)$ (3-space) and $\beta\big((q+1)\dot a_4 + (p-\frac{1}{2})\dot b_4/b_4\big)$ (extra times). Keeping the warp of the extra times ($\beta = 1/6$) forces $\alpha = 0$, $q = -1$, $p = 1/2$: the record's current metric, used by UM-2,

$$\bar g = \mathrm{diag}\big(e^{-2a_4}b_4\cot^2 z,\ e^{2a_4},\ e^{2a_4},\ e^{2a_4},\ -1,\ -e^{-2a_4}b_4(\sin^2 z)^{1/6},\ -e^{-2a_4}b_4(\sin^2 z)^{1/6},\ -e^{-2a_4}b_4(\sin^2 z)^{1/6}\big).$$

Every off-diagonal component of $E_{(1)}, E_{(2)}, E_{(3)}$ vanishes identically for all $a_4(x_4)$, $b_4(x_4) > 0$ and every real z (record: sympy 14/14, Wolfram 13/13; re-derived in UM-2, section 8). The alternative K of the record ($g_{00} = e^{2a_4}\cot^2 z$, $g_{ii} = e^{2a_4}(\sin^2 z)^{1/6}$, $g_{tt} = -e^{-2a_4}b_4$) is also diagonal at every order; the record leaves the choice to the author, and UM-2 uses version 2.

5.2 The hidden coordinate, the fold and the regular tip

On every interval with $\sin z \neq 0$,

$$y = \frac{\ln|\sin z|}{6H} \in (-\infty, 0], \qquad dy = \cot z \, dx_0, \qquad f = e^{-a_4} b_4^{1/2},$$

$$ds^2 = e^{2a_4} d\vec{x}^{\,2} - dx_4^2 + f^2 \big[dy^2 - e^{2Hy}(dx_5^2 + dx_6^2 + dx_7^2) \big].$$

The fold is $z = \pi/2$ ($y = 0$); the tips are $z \to 0^+$ and $z \to \pi^-$ ($y \to -\infty$). The record replaces the tip by a regular cutoff at $y = -L$ with

$$\eta \equiv HL = 3,$$

a chosen cutoff (the extended levels depend on it like those of a box of proper length fL). UM-2 keeps $\eta = 3$ (ASSUMED).

5.3 The dirac16complex Lagrangian and the canonical spin connection

The specification of the record (under the author's orders) fixes the Dirac-type Lagrangian

$$\mathcal{L}_\Psi = \sqrt{|g|} \Big[\tfrac{1}{2} \big(\bar\Psi \gamma^M D_M \Psi - (D_M \bar\Psi) \gamma^M \Psi \big) - m \, \bar\Psi \Psi - U(S) \Big], \qquad S = \bar\Psi \Psi, \quad U(S) = \frac{\lambda}{2} S^2, \quad \bar\Psi = \Psi^\dagger C,$$

$$D_M \Psi = \partial_M \Psi + \Omega_M \Psi, \quad \Omega_M = \tfrac{1}{2} \omega_{Mab} S^{ab}, \quad \omega_M{}^a{}_b = e^a{}_N \big(\partial_M e_b{}^N + \Gamma^N{}_{ML} e_b{}^L \big), \quad \gamma^M = e_a{}^M \gamma^a,$$

the canonical (vielbein-postulate, torsion-free) spin connection. The record chose the Dirac-type form because the notebook's real Majorana-type Lagrangian is a total derivative for anticommuting real fields (it has no field equations), and it uses the canonical connection rather than the notebook's contraction of the mixed $\omega_M{}^a{}_b$, which lacks a metric factor. The field equation is $\gamma^M D_M \Psi = (m + U'(S))\Psi$.

5.4 Canonical quantisation and the Krein metric

With x_4 as the evolution time, $\{\Psi_A(x), \Psi_B^\dagger(x')\} = B_{AB} \, \delta^7(x - x')/\sqrt{|g|}$ on a slice $x_4 = \text{const}$; B has eigenvalues $+1$ (8 times) and -1 (8 times), so the inner product is indefinite (a Krein space). The record works in the good sector (no extra-time momentum), counts the positive-energy branch of every block as particles, and evaluates expectation values as $\langle \Psi^\dagger M \Psi \rangle = u^\dagger B M u$.

5.5 The Kohn-Sham reduction

For the corrected metric the record reduces the 16-component equation, with $\Psi = e^{ik\cdot x} \, e^{-\frac{3}{2}Hy} \, e^{a_4/2} b_4^{-1} \, V_\beta \, \chi(y, x_4)$, to the 2×2 block Hamiltonians

$$h_j = j \Big[-\frac{i}{f} \sigma_1 \partial_y + M(y) \, \sigma_2 + \kappa k \, \sigma_3 \Big] + v(y), \qquad \kappa = e^{-a_4},$$

in the blocks (j, s_2, s_3) of the commuting set $J = \gamma^{x_0} \gamma^{x_1} \gamma^{x_4}$, $K_1 = \gamma^{x_2} \gamma^{x_3}$, $K_2 = \gamma^{x_5} \gamma^{x_6}$ (eigenvalues j, is_2, is_3), with the contact-interaction mean fields of the record's Hartree-plus-exchange treatment of the contact interaction

$$e_{int} = \frac{15}{32} \lambda S^2 - \frac{1}{32} \lambda n^2, \qquad M_{eff} = m + \frac{15}{16} \lambda S, \qquad v = -\frac{\lambda}{16} n,$$

the fold conditions $\theta \in \{0, \pi\}$ (both parities filled, weight $w = 1/2$ each), the regular tip, and the densities carrying $P = e^{-3Hy}/\mathrm{Vol}_7$. The record's mass is $m = 3HM$ with $M = 1/3$, i.e. $m = H$; UM-2 keeps it (ASSUMED). Two reference states of the record (at $f = 1$, $a_4 = 0$: $N = 8$ with $\lambda = 157.7$, and $N = 592$ with $\lambda = 2.131$) serve as validation targets of the independent UM-2 solver (section 8).

5.6 Earlier work used only for comparison

The first universe model of this repository (UM-1, folder `Revision/universe_model`) is not an input of UM-2. UM-2 differs from it in keeping the author's weights and Λ, in the electroweak sector (both halves of Ψ, the chiral tip Z portal, N_{eff}), and in the dark-matter candidate.

6 Standard physics and measured values (STANDARD)

Every measured value enters UM-2 only as the target of a fit or as a comparison, never as a derived result.

input	value used	source	status
W mass	80.3625 ± 0.0077 GeV	PDG 2026, OUR EVALUATION (pdgLive S043M)	verified
Z mass	91.1879 ± 0.0020 GeV	PDG 2026, OUR AVERAGE (pdgLive S044M)	verified
$\sin^2\theta_W$ ($\overline{MS}$, M_Z)	0.23122 ± 0.00004	PDG electroweak review	from memory
electroweak ρ parameter	$\rho_0 - 1 = 0.00038 \pm 0.00020$	PDG electroweak review (global fit)	from memory
electron mass	0.51099895 MeV	PDG, CODATA	from memory
u, d masses ($\overline{MS}$, 2 GeV)	2.16 ± 0.07 MeV, 4.70 ± 0.07 MeV	PDG	from memory
neutrino mass of the one generation	0.05 eV $\approx \sqrt{\lvert\Delta m^2_{31}\rvert}$	chosen	ASSUMED
Fermi constant, electroweak scale	$G_F = 1.1663788 \times 10^{-5}$ GeV^{-2}, $v = 246.22$ GeV	PDG	from memory
fine-structure constant	$\alpha^{-1}(0) = 137.035999$	CODATA	from memory
invisible Z width	$N_\nu = 2.9840 \pm 0.0082$	LEP/SLD, Phys. Rept. 427 (2006) 257	from memory
Z widths	$\Gamma_Z = 2495.5$ MeV, $\Gamma_{inv} = 499.0$ MeV	PDG	from memory
nucleon masses	$m_p = 938.272$ MeV, $m_n = 939.565$ MeV	PDG	from memory
$\hbar c$, k_B	$1.973269804 \times 10^{-16}$ GeV m, $8.617333262 \times 10^{-5}$ eV/K	CODATA	from memory
CMB temperature	$T_0 = 2.7255$ K	Fixsen 2009	from memory
three-flavour N_{eff}	3.0440	Bennett et al. 2021	from memory
neutral-current neutrino decoupling	$T \approx 2.6$ MeV	standard estimate	from memory
Planck 2018 base parameters	$\omega_b = 0.02237$, $\omega_c = 0.1200$, $100\,\theta_{MC} = 1.04092$, $\tau = 0.0544$, $\ln(10^{10} A_s) = 3.044$, $n_s = 0.9649$	Planck 2018 results VI, Table 2	from memory
N_{eff} with free N_{eff}	2.99 ± 0.17	Planck 2018 results VI	from memory
BAO	DESI DR2: 13 measurements of D_V/r_d, D_M/r_d, D_H/r_d (table below)	DESI collaboration, arXiv:2503.14738	values verified (2026-10-17) against Table 2 of arXiv:2603.26861; correlations from memory
dark-matter density	$\Omega_c h^2 = 0.120$	Planck 2018	from memory
Planck mass	$M_{Pl} = 1.22 \times 10^{19}$ GeV		from memory
direct-detection limit	LZ: 2.2×10^{-48} cm^2 at 40 GeV, scaled linearly with the mass	LZ 2024	from memory; the extrapolation is an ESTIMATE
nucleon gluon content, strong coupling	$f_{TG} = 0.92$, $\alpha_s = 0.3$ (low scale)	standard	from memory; ESTIMATE

input	value used	source	status
dark-energy density	$\rho_\Lambda \approx 2.5\times10^{-47}$ GeV^4		from memory
supernova dark-energy fits	constant $w=-0.764$; $(w_0,w_a)=(-0.861,-0.60)$	the author's private file (not in the repository)	AUTHOR

The DESI DR2 BAO measurements used (z; D_V/r_d; D_M/r_d; D_H/r_d; correlation r_{MH} of D_M and D_H):

tracer	z	D_V/r_d	D_M/r_d	D_H/r_d	r_{MH}
BGS	0.295	7.942 ± 0.075			
LRG1	0.510		13.588 ± 0.167	21.863 ± 0.425	-0.459
LRG2	0.706		17.351 ± 0.177	19.455 ± 0.330	-0.404
LRG3+ELG1	0.934		21.576 ± 0.152	17.641 ± 0.193	-0.416
ELG2	1.321		27.601 ± 0.318	14.176 ± 0.221	-0.434
QSO	1.484		30.512 ± 0.760	12.817 ± 0.516	-0.500
Lyα	2.330		38.988 ± 0.531	8.632 ± 0.101	-0.431

with $\chi^2=\sum_i (d_i-t_i)^T C_i^{-1}(d_i-t_i)$, $C_i=\begin{pmatrix}\sigma_M^2 & r_{MH}\sigma_M\sigma_H\\ r_{MH}\sigma_M\sigma_H & \sigma_H^2\end{pmatrix}$ for the anisotropic tracers and $(d-t)^2/\sigma_V^2$ for BGS, where $D_M(z)=\int_0^z c\,dz'/H(z')$ (flat), $D_H=c/H(z)$, $D_V=(zD_M^2D_H)^{1/3}$.

Standard theory used: the Standard-Model gauge theory $SU(3)_c\times SU(2)_L\times U(1)_Y$ with one generation; QCD confinement of u, d into protons and neutrons; the tree-level Z partial width $\Gamma(Z\to f\bar f)=N_c\,G_F m_Z^3(g_V^2+g_A^2)/(6\sqrt{2}\pi)$, $g_V=T_3-2Q\sin^2\theta_W$, $g_A=T_3$; the freeze-out of a relic (Kolb and Turner); the trace-anomaly nucleon matrix element; the Einstein-Boltzmann equations of linear cosmological perturbations (solved by CAMB); Higgsless electroweak symmetry breaking by a non-linear sigma field and fermion localisation in an extra dimension (concepts only, no numbers).

7 Ingredients new in UM-2 (ASSUMED and INVENTED)

7.1 The metric as a power series (ASSUMED)

$$g_{MN}=\bar g_{MN}+\epsilon\,h^{(1)}_{MN}+\epsilon^2 h^{(2)}_{MN}+\dots,\qquad \epsilon\sim10^{-5}\ \text{(the CMB anisotropy)},$$
$$h^{(1)}_{MN}(x^\mu,x_0)=\sum_{n\ge0} h^{(1,n)}_{MN}(x^\mu)\,u_n(x_0),$$

with u_n the eigenfunctions of the x_0 operator; the flat $n=0$ term in the Newtonian gauge,

$$ds^2=e^{2a_4}\big[(1-2\epsilon\Psi_N)\delta_{ij}+\epsilon h^{TT}_{ij}\big]dx_i dx_j-(1+2\epsilon\Phi)dx_4^2+f^2(1+2\epsilon\sigma)\big[dy^2-e^{2Hy}d\vec{x}_t^{\,2}\big],$$

carries the CMB, CNB and BAO physics; the radion σ and the $n\ge1$ Kaluza-Klein modes are massive.

7.2 The block stabiliser (ASSUMED; its origin is OPEN)

$$S_{stab}=-\int d^8x\sqrt{|g|}\;\rho_b(f),\qquad \kappa\rho_b(f)=\kappa\rho_b^*\Big(\frac{f_*}{f}\Big)^8,$$
$$T^M{}_N[\text{stab}]=\text{diag}(p_0,p_3,p_3,p_3,-\rho_b,p_t,p_t,p_t),\qquad p_3=-\rho_b,\qquad p_0=p_t=\rho_b.$$

It is conserved ($\dot\rho_b=-3A(\rho_b+p_3)-4h(\rho_b+p_0)=-8h\rho_b$, so $\rho_b\propto f^{-8}$), four-dimensionally Lorentz invariant ($p_3=-\rho_b$) and isotropic in the block ($p_0=p_t$). It was introduced because the author's weights and Λ admit no static block with attractive four-dimensional gravity without a source: the only sourceless static block with $A=0$ is $k_0=1/(4w_2)=8$, where the coefficient of the four-dimensional Einstein-Hilbert term is $G(k_0)=1-24w_2k_0+72w_3k_0^2=-1$ (repulsive), with $\Lambda_0=-24H^2\neq-10H^2$ (`um2_background.py`). The family $p_0=p_t=\nu\rho_b$, $\rho_b\propto f^{-4(1+\nu)}$, $\nu\in[1/4,3]$, was also computed.

7.3 The Standard-Model fields in eight dimensions (ASSUMED structure; boundary conditions INVENTED)

$$S_{gauge} = -\frac{1}{4}\int d^8x\sqrt{|g|}\,g^{MP}g^{NQ}\Big(\frac{1}{\hat g_s^2}G^\alpha_{MN}G^\alpha_{PQ}+\frac{1}{\hat g^2}W^a_{MN}W^a_{PQ}+\frac{1}{\hat g'^2}B_{MN}B_{PQ}\Big),$$

$$S_f=\sum_\psi\int d^8x\sqrt{|g|}\Big[\tfrac12\big(\bar\psi\gamma^M D_M\psi-(D_M\bar\psi)\gamma^M\psi\big)-\mu_\psi\bar\psi\psi\Big],\qquad D_M=\partial_M+\Omega_M-iG^\alpha_M T^\alpha-iW^a_M\frac{\tau^a}{2}-iYB_M,$$

for one generation $Q=(u,d)_L$, u_R, d_R, $L=(\nu,e)_L$, e_R, ν_R (hypercharges $1/6, 2/3, -1/3, -1/2, -1, 0$), each a 16-component spinor of the same Clifford algebra times its gauge indices. With the selection projector

$$P_{sel}=\frac{1+\Gamma_4\gamma^{x_0}}{2}\,\frac{1-i\gamma^{x_5}\gamma^{x_6}}{2},\qquad \Gamma_4=i\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4},$$

the conditions at the fold and the tip are, for doublets ($\mu_\psi>0$), $P_{sel}(1-\gamma^{x_0})\psi=0$, $(1-P_{sel})(1+\gamma^{x_0})\psi=0$ at the fold and $(1-\gamma^{x_0})\psi=0$ at the tip, and for singlets ($\mu_\psi<0$) the same with $\gamma^{x_0}\to-\gamma^{x_0}$. The weak bosons are $W^\pm=(W^1\mp iW^2)/\sqrt2$, with electric charges $+e$ (W^+) and $-e$ (W^-).

7.4 The tip sigma field (INVENTED for this geometry; no Higgs particle)

$$S_{tip}=\int_{y=-L}d^7x\sqrt{|g_{tip}|}\Big[-\frac{v_\Sigma^2}{4}\mathrm{tr}\big((D_\mu\Sigma)^\dagger D^\mu\Sigma\big)-\big(\bar L\,\Sigma\,\Lambda_\ell\,\ell_R+\bar Q\,\Sigma\,\Lambda_q\,q_R+\mathrm{h.c.}\big)\Big],\qquad \Sigma\in SU(2),$$

$$D_\mu\Sigma=\partial_\mu\Sigma-iW^a_\mu\frac{\tau^a}{2}\Sigma+iB_\mu\Sigma\frac{\tau^3}{2},\qquad \Sigma\to U_L\Sigma U_Y^\dagger,\quad U_Y=e^{i\theta\tau^3/2}.$$

7.5 The portal interactions (INVENTED)

$$S_{I9}=\int_{y=-L}d^7x\sqrt{|g_{tip}|}\big[c_Z+c'_Z\varsigma\big](i\bar\Psi\gamma^\mu P_+\Psi)\,\frac{iv_\Sigma^2}{2}\mathrm{tr}\big(\Sigma^\dagger D_\mu\Sigma\,\tau^3\big),\qquad P_+=\frac{1+\Gamma_4}{2},\quad \varsigma=\pm1,$$

$$S_{I11}=\int d^8x\sqrt{|g|}\,\frac{c_V}{\Lambda_\star^6}(i\bar\Psi\gamma^M\Psi)\sum_\psi q_\psi(i\bar\psi\gamma_M\psi),$$

$$S_{I12}=-\int d^8x\sqrt{|g|}\,\frac{\bar\Psi\Psi}{\Lambda_\star^3}\big[c_GG^\alpha_{MN}G^{\alpha MN}+c_WW^a_{MN}W^{aMN}+c_BB_{MN}B^{MN}\big].$$

Operators considered and rejected: a dipole portal $\bar\Psi S^{MN}\Psi B_{MN}$ (it would thermalise the sterile Ψ_B fold band through the photon: too much dark radiation) and any operator coupling the neutral Ψ to a single W^+ or W^- (none is gauge invariant).

7.6 Parameter choices (ASSUMED)

$\eta=3$ and $m=H$ (from the record); the tip mixing $\kappa_f=1$ for every fermion with $c_L=-c_R$; the neutrino mass 0.05 eV; $c_V=0$ (benchmark); the Kohn-Sham benchmark self-coupling $\lambda_B=7\times10^{-5}H^{-6}$ (mean fields below 5 % of m in the densest benchmark state); the extra-time coordinate volume v_t (it enters only $V_\perp$).

8 The computations and their verification (DERIVED)

All computations are Python scripts in the folder `universe_model_2/code`; each writes its numbers to `universe_model_2/outputs` and a report of named checks (name, verdict, detail) to `universe_model_2/reports`; no single run exceeds 423 s. In total 177 checks, none failed.

script	what it computes	method	checks
`um2_algebra.py`	the author's gamma matrices rebuilt from the notebook formulas; C, Γ, B; Pin(4,4) irreducibility; the two inequivalent Spin(4,4) halves; the 2×2 block basis and the block forms of every matrix the reductions use	sympy, exact	31/31
`um2_geometry.py`	Riemann tensor of $\bar{g}$ in two charts; Lovelock tensors by four routes (pairing formula, Ricci, explicit Gauss-Bonnet tensor, minisuperspace variation with a lapse); Bianchi identities; the record's equations; the static block; the canonical spin connection	sympy, exact	25/25
`um2_background.py`	the class-form background equations; the static block; the stabiliser and its family; the radion; the linear response of the block to matter sources; the dimensional reduction	sympy, exact; eigenvalues numerically	22/22
`um2_fields.py`	the exact x_0 reduction of the eight-dimensional Dirac equation (all blocks, general a_4, b_4, $M(y)$, $v(y)$); boundary conditions; zero modes; tip mixing; the gauge x_0 operator; gauge invariance of the tip terms	sympy and numpy	15/15
`um2_spectra.py`	the x_0 spectra of gauge bosons and graviton; the W, Z fit to ρ_0; the fermion masses as exact x_0 eigenvalues	mpmath root finding; quadratic finite elements with Richardson; ODE shooting	13/13
`um2_ks_jobs.py`	Kohn-Sham: validation against the record (2/2), free spectrum (3/3), interacting ground states $N = 8$ (3/3) and $N = 592$ (4/4), dark-matter quanta (3/3), portal responses (5/5), block feedback (2/2)	Galerkin in the free basis, safeguarded Anderson self-consistency, Richardson in the basis size	22/22

script	what it computes	method	checks
`um2_electroweak.py`	charges (W^+: $+e$, W^-: $-e$), anomalies, couplings, P_+, $\Gamma_4 = -B\gamma^{x_0}$, Z charges of Ψ_A, Ψ_B, N_ν, N_{eff}, CNB, Z widths	sympy and numerics	19/19
`um2_primordial.py`	the nonlinear primordial evolution, the basin, the separatrix, the radion oscillation	DOP853 integration, bisection	5/5
`um2_cosmology.py`	TT and EE spectra, BAO χ^2, N_{eff}, helium, the variant with a thermalised Ψ_B fold band	CAMB 2.0.5	5/5
`um2_analysis.py`	the x_0 decompositions and the uncertainty relation; the gap theorem; sum rules; relic abundance; direct detection; the condensate; the vacuum energy; the sensitivities to the block size	mpmath, numpy	20/20

8.1 The principal numbers and where they come from

quantity	value	computed by
static block k_*, f_*, $G(k_*)$, $\kappa\rho_b^*$	$12 + \frac{4}{3}\sqrt{51} = 21.5219$, 0.215556, $\frac{20}{3} + \sqrt{51} = 13.8081$, $(26 + 4\sqrt{51})H^2$	`um2_background.py`
radion	$\omega_r^2/H^2 = 31552/2249 - 2416\sqrt{51}/6747 = 11.4721$, $m_r = 3.387H = 48.38$ TeV	`um2_background.py`, `um2_spectra.py`
unbroken x_0 spectrum ($\eta = 3$)	$\Lambda = 0$; $\Lambda_1 = 30.1196$; odd 27.0890; gap 20.25	`um2_spectra.py`
$W^\pm$, Z	$\beta_W = 0.011932$, $\beta_Z = 0.015364$, $\Lambda_W = 1.32374 \times 10^{-5}$, $\Lambda_Z = 1.70374 \times 10^{-5}$; Dirichlet bound $\Lambda_D = 0.010016$	`um2_spectra.py`
scales	$m_{KK} = 121.22$ TeV, $H = 14.283$ TeV, proper half cell $fL = 8.934 \times 10^{-21}$ m, x_0 period 7.234×10^{-21} m, gap 99.40 TeV	`um2_spectra.py`
fermion localisations ($\kappa_f = 1$)	$c_\nu = 18.6709$, $c_e = 10.3038$, $c_u = 9.5448$, $c_d = 9.1341$	`um2_spectra.py`
dirac16complex at f_*	$c_\psi = 0.646667$; $\varepsilon_{DM} = 3.102204\,m$ ($p = 2.93661\,m$); even bulk $4.959981\,m$; x_0-momentum fraction 0.896	`um2_ks_jobs.py spectrum`
dark matter	$M_{DM} = 44.31$ TeV; $w = 0$; $p_0/\rho = 0.772034343$ $= -d\ln\varepsilon_{DM}/d\ln f$	`um2_ks_jobs.py darkmatter`

quantity	value	computed by
Z portal	$r^2 = 0.9920 \pm 0.0041$; Ψ_A decoupling at 2.61 MeV; $N_{eff} = 3.044$; CNB 1.9454 K, 112.0 per cm^3 per species, 336 in all	`um2_electroweak.py`
one-generation Z width	1232 MeV (measured 2495.5 MeV)	`um2_electroweak.py`
primordial era	separatrix $k_c = 2.6983, 2.5991, 2.5301$; deflation factor up to 2.9166; radion $3.3868H$ measured	`um2_primordial.py`
cosmology	$H_0 = 67.43$ km/s/Mpc, $r_{drag} = 147.105$ Mpc, BAO $\chi^2 = 30.31$ (13 points; ΛCDM 32.55), TT/EE within $2.2 \times 10^{-4}/3.7 \times 10^{-4}$ of ΛCDM	`um2_cosmology.py`
relic	$x_f = 30.31$, $g_* = 56.75$, $\langle\sigma v\rangle = 1.37 \times 10^{-25}$ cm^3/s, $(8G_g^2 + 3G_W^2 + G_B^2)^{1/2} =$ (16.6 TeV)$^{-3}$	`um2_analysis.py`
direct detection	$\sigma_{SI} = 4.3 \times 10^{-53}$ cm^2 (gluon-only) against 2.4×10^{-45} cm^2	`um2_analysis.py`
dark energy	condensate tuned to $w_0 = -1$: $w_a = -6$, energy zero at $z = 0.260$; vacuum-energy ratio 5.5×10^{64}; $(H_0/H)^2 = 1.0 \times 10^{-92}$	`um2_analysis.py`
uncertainty relation	m/m_H: first KK 7.49, first odd 6.47, dark matter 2.29, W 1.14×10^{-3}, Z 1.29×10^{-3}, e 3.2×10^{-9}, ν 1.7×10^{-16}	`um2_analysis.py`
sensitivities $d\ln X/d\ln f$	KK -1, m_W -0.501, m_e -20.6, m_u -19.1, m_d -18.3, m_ν -37.3, M_{DM} -0.772, G_N and α -4	`um2_analysis.py`

8.2 Methods, with the equations they use

- **Exact checks.** Symbolic identities are verified with sympy by simplifying the difference to zero; matrix identities entry by entry.
- **Boson eigenvalues.** With $t = y/L$, $u = e^{-3\eta t/2}w$, $q^2 = \Lambda - 9\eta^2/4$, the operator $-e^{-3\eta t}(e^{3\eta t}u')' = \Lambda u$ with $u'(0) = 0$ (even) or $u(0) = 0$ (odd) and $u'(-1) = \beta u(-1)$ has the eigenvalue conditions $\left[q^2 + \frac{3\eta}{2}(\beta + \frac{3\eta}{2})\right]\sin q - \beta q\cos q = 0$ (even) and $q\cos q + (\beta + \frac{3\eta}{2})\sin q = 0$ (odd), solved with mpmath at 30 digits. Cross-check: quadratic finite elements for $\int e^{3\eta t}u'^2dt + \beta e^{-3\eta}u(-1)^2 = \Lambda\int e^{3\eta t}u^2dt$ on 100, 200, 400 elements, the eigenvalue taken from the Rayleigh quotient of the gradient form and extrapolated by Richardson.
- **Richardson extrapolation.** For a quantity with error $C\,n^{-p}$: $x \approx x_{2n} + (x_{2n} - x_n)/(2^p - 1)$; the observed order is $\log_2[(x_n - x_{2n})/(x_{2n} - x_{4n})]$.
- **Fermion eigenvalues.** The exact condition $D(e) = e^2 s_L s_R - \kappa_f^2(C_L - c_L s_L)(C_R + c_R s_R)$, $s = \sinh w/w$, $C = \cosh w$, $w = \sqrt{c^2 - e^2}$, solved with mpmath; the first-order solution $e = \kappa_f\zeta(c_L)\zeta(c_R)$, $\zeta(c)^2 = c\,e^{-c}/\sinh c$; cross-check by direct integration of the coupled (L, R) Dirac system (DOP853, relative tolerance 10^{-13}).
- **Kohn-Sham.** Orbitals of h_j are expanded in the exact eigenfunctions of the free problem of each sector (60, 120 or 240 functions, quadrature with ten points per function), the mean fields $M_{eff} = m + \frac{15}{16}\lambda S$, $v = -\frac{1}{16}\lambda n$ are iterated to self-consistency (tolerance 10^{-13}) with a safeguarded Anderson mixing and a fixed labelling of the occupied set; checks: the energy identity $E_{KS} = 2\,\mathrm{Vol}_7\int_{-L}^{0} e^{3Hy}\rho\,dy$, the y-conservation law $\left[e^{3Hy}p_0\right]_{-L}^{0} = 3H\int_{-L}^{0} e^{3Hy}p_t\,dy$, Richardson in the basis size, the Hellmann-Feynman theorem $d\varepsilon/de =$

$\langle\chi|\partial h/\partial e|\chi\rangle$ for every portal mean field, and the reproduction of the record's two reference states to 10^{-13} (energies, HOMO, LUMO) and 10^{-9} (integrated hidden pressure).

- **Primordial era.** The 3-space and block components of the field equations are solved for $\dot{A}$, $\dot{h}$, with $\dot{k} = -2kh$ and the stabiliser $\kappa\rho_b \propto k^4$; the x_4 constraint fixes the initial A and is monitored (residual below 6×10^{-10}); the separatrix is found by bisection in the initial k.
- **Cosmology.** CAMB 2.0.5 with the Planck 2018 parameters above, θ_{MC} fixed (so H_0 is an output), one massive species (ν_e, 0.05 eV) and $N_{eff} = 3.044$ in total, $\ell_{max} = 2600$, lensing included; the comparison model has three neutrinos with $\sum m_\nu = 0.06$ eV.
- **Relic abundance.** For $\mathcal{L} = G\,\bar{\chi}\chi F_{\mu\nu}F^{\mu\nu}$ per abelian channel, $\sigma v = G^2 s(s-4M^2)/(4\pi) \to G^2M^4v^2/\pi$ (p-wave); in total $\sigma v = (M^4v^2/\pi)(8G_g^2 + 3G_W^2 + G_B^2)$, $\langle\sigma v\rangle(x) = 6b/x$ with $b = (M^4/\pi)(8G_g^2 + 3G_W^2 + G_B^2)$; freeze-out (Kolb and Turner, Dirac fermion, $g = 2$, $c = 1/2$): $x_f = \ln\left[0.038\,c(c+2)\,g\,M_{Pl}M\langle\sigma v\rangle(x_f)/\sqrt{g_* x_f}\right]$, $\Omega h^2 = 2 \times 1.07 \times 10^9\,\mathrm{GeV}^{-1}\,x_f/\left(\sqrt{g_*}\,M_{Pl}\,3b/x_f\right) = 0.120$, with $g_* = 56.75$ (one generation without ν_R, the two Ψ_A species, the gauge bosons; no Higgs).
- **Direct detection** (ESTIMATE). Trace anomaly: $\langle N|G^\alpha G^\alpha|N\rangle = -\frac{8\pi}{9\alpha_s}m_N f_{TG}$, $f_N = G_g\frac{8\pi}{9\alpha_s}m_N f_{TG}$, $\sigma_{SI} = \mu^2 f_N^2/\pi$.
- **Uncertainty relation.** For bosons, with $u = e^{-3\eta t/2}v$: $e^{3\eta t}u'^2 = v'^2 + \frac{9}{4}\eta^2 v^2 - \frac{3}{2}\eta(v^2)'$, hence $(m\,fL)^2 = \langle p^2\rangle + \frac{9\eta^2}{4} - \frac{3\eta}{2}\rho(\mathrm{fold}) + (\frac{3\eta}{2} + \beta)\rho(\mathrm{tip})$; the Heisenberg estimate is $m_H = \hbar c/(2\Delta s)$ with Δs the proper spread of the profile in the hidden direction.

8.3 Corrections made during the work

1. The gauge-invariance test of the tip terms first used the wrong sign of the hypercharge phase; with $\psi \to e^{iY\theta}\psi$ every term is invariant to 4×10^{-15}.
2. The finite-element cross-check of Λ_W was first limited by round-off; the gradient-form Rayleigh quotient with Richardson extrapolation (observed order 4.01) agrees with the exact value to 3×10^{-14}.
3. The Kohn-Sham basis convergence and the conservation law were first checked without extrapolation; Richardson in the basis size (observed orders 4.95 for the energy, 2.97 for the hidden pressure) was added.
4. The self-consistency loop did not converge for $N = 592$ at the record's coupling; a fixed labelling of the occupied set and a safeguarded Anderson mixing were introduced, the benchmark coupling λ_B was adopted, and the record's coupling is kept as a stress test, which now converges.
5. A first statement that every primordial start above the unstable static block reaches the stable one was false in detail; the separatrix $k_c(h_0)$ was computed by bisection instead.
6. The primordial figure was first sampled too coarsely (aliasing of the radion oscillation) and is now densely sampled.
7. The sum rules of the uncertainty analysis were first imprecise for the sharp fermion zero modes; exact mpmath normalisation and the tip term fixed them (10^{-8}).
8. The radion quantity was stored under a wrong sign label; it is now stored as $\omega_r^2 = \det$ of the block Jacobian.
9. Portal names of the first universe model (P1 to P6) in two check texts were replaced by I9, I11, I12.
10. The gauge-kinetic portal was first written with the scale to the power 5, which is dimensionally wrong in the geometric normalisation of the gauge fields; it is $\bar{\Psi}\Psi\,F_{MN}F^{MN}/\Lambda_\star^3$, and the symbol of the portal scale was changed so that it no longer clashes with the Dirichlet bound Λ_D.
11. Before publication every number of the documents was compared with the computed outputs; four statements were corrected (the dark-matter momentum $p\,fL = 1.89901$, the finite-element agreement 5×10^{-10}, the TT and EE agreements 2.2×10^{-4} and 3.7×10^{-4}, and the restriction $A = 0$ of the sourceless static block).
12. The kinetic term of the tip sigma field was first written with the sign of the signature $(+,-,-,-)$; in the signature $(+,+,+,-)$ of $x_1,\ldots,x_4$ it is $-\frac{v_\Sigma^2}{4}\mathrm{tr}((D_\mu\Sigma)^\dagger D^\mu\Sigma)$, which gives the mass terms $-\frac{1}{2}m^2V_\mu V^\mu$ with $m^2 > 0$ (the computations always used $m^2 > 0$).
13. With the real matrices $C\gamma^a$ antisymmetric, $\bar{\Psi}\gamma^a\Psi$ is anti-Hermitian; the portal currents are now written as the Hermitian currents $i\bar{\Psi}\gamma^M\Psi$ (I9, I11) and the Standard-Model currents as $i\bar{\psi}\gamma^M\psi$.
14. The block response to the hidden pressure of a Kohn-Sham state first used p_0 alone; the common block scale is driven by the block-averaged pressure $\bar{p} = (p_0 + 3p_t)/4$, and $p_0 - p_t$ drives a block-shape mode outside the order-0 metric (OPEN). The linearised sourced equations were added (`um2_background.py`, 2 checks).

8.4 What is not in the repository, and why

By the author's instruction the following private inputs are never committed (they are git-ignored): the prompt files, the author's private PDF with the supernova dark-energy fits, the folders `dirac-main/` and `vendor/`, the archive `dirac-main_2.zip`, the files `Generalized_Kronecker_Delta.*`, and the notebook `WaveFunctionUniverse-4+4-Einstein-Lovelock.nb` with its `.mx` files. UM-2 uses from them only what is written out in this document (the notebook formulas of the metric and of the gamma matrices, and the two supernova fits).

8.5 Reproduction

From the repository root, with Python 3 and the packages sympy, numpy, scipy, mpmath, matplotlib and camb installed (`pip install sympy numpy scipy mpmath matplotlib camb`), the commands

```
python universe_model_2/code/um2_algebra.py
python universe_model_2/code/um2_geometry.py
python universe_model_2/code/um2_background.py
python universe_model_2/code/um2_fields.py
python universe_model_2/code/um2_spectra.py
python universe_model_2/code/um2_ks_jobs.py validate
python universe_model_2/code/um2_ks_jobs.py spectrum
python universe_model_2/code/um2_ks_jobs.py ground8
python universe_model_2/code/um2_ks_jobs.py ground592
python universe_model_2/code/um2_ks_jobs.py darkmatter
python universe_model_2/code/um2_ks_jobs.py portals
python universe_model_2/code/um2_ks_jobs.py feedback
python universe_model_2/code/um2_electroweak.py
python universe_model_2/code/um2_primordial.py
python universe_model_2/code/um2_cosmology.py
python universe_model_2/code/um2_analysis.py
python universe_model_2/tools/build_docs.py
```

rerun every computation in dependency order (each exits with code 0 if and only if all its checks pass) and rebuild the PDF documents from the Markdown sources (pandoc and a TeX Live installation with the `lmodern` package are needed for the last command). A rerun from a fresh clone of the repository (2026-10-17) reproduced every stored number to a relative 10^{-9} or better; quantities at the round-off level (10^{-14}, e.g. HOMO deviations) differ in their last digits because the threaded linear algebra does not sum in a fixed order. The document build is deterministic: no dates in the LaTeX sources, PDF dates and trailer identifiers suppressed, a fixed `SOURCE_DATE_EPOCH`.

Software used: Python 3.13.16, sympy 1.14.0, numpy 2.5.3, scipy 1.18.1, mpmath 1.3.0, matplotlib 3.11.2, camb 2.0.5, pandoc 3.1.3, pdfTeX 1.40.25 (TeX Live 2023).

8.6 History in the repository

UM-2 was started on top of commit `0cb4916`. Its computations were committed in the order algebra and geometry (`ca523d9`), background gravity (`0b3e10d`), field reductions (`6fc35a0`), mass spectra (`dcfd528`), Kohn-Sham (`2f3018e`), electroweak sector (`2e4c48d`), primordial era (`a502e62`), cosmology (`717b46e`), analysis (`a06b55e`), followed by the corrections listed above and the documents.